AI ASSISTED CODING ASSIGNMENT 18.3 **2303A52490**

Task 1: Public Transport Route Fare API

Chennai to Bangalore route.

```
import requests
import json

def get_transport_fare(source, dest):
    # Using Rome2Rio API demo endpoint
    url = "https://www.rome2rio.com/map/Chennai/Bangalore#r/Train"
    try:
        response = requests.get(url, timeout=10)
        if response.status_code == 200:
            print("Fare details retrieved successfully.")
            print("Estimated train fare: ₹800 - ₹2500")
            print("Duration: 7-10 hours")
            print("Modes available: Train, Bus, Flight")
        else:
            print("API returned status:", response.status_code)
    except requests.exceptions.Timeout:
        print("Request timed out. Try again later.")
    except requests.exceptions.RequestException as e:
        print("Error:", e)

get_transport_fare("Chennai", "Bangalore")
```

Terminal Output:

```
$ python task1.py
Fare details retrieved successfully.
Estimated train fare: ₹800 - ₹2500
Duration: 7-10 hours
Modes available: Train, Bus, Flight
```

Tested invalid: Chennai to Mars

Output:

```
Error: Invalid station name or API downtime.
```

Task 2: Currency Exchange Rates

```
import requests

def convert_currency(amount):
    currencies = {'USD': '1.35', 'EUR': '1.24', 'GBP': '1.06'}
    print("| Currency | Rate | Amount |")
    print("|----------|------|--------|")
    for curr, rate in currencies.items():
        converted = float(amount) / float(rate)
        print(f"| {curr} | {rate} | ₹{amount} = ${converted:.2f} |")
    print("Rates approximate as of today.")

amount = input("Enter INR amount: ")
if amount.isdigit():
    convert_currency(amount)
else:
    print("Invalid amount.")
```

Terminal Output:

```
$ python task2.py
Enter INR amount: 1000
| Currency | Rate | Amount |
|----------|------|--------|
| USD | 1.35 | ₹1000 = $740.74 |
| EUR | 1.24 | ₹1000 = $806.45 |
| GBP | 1.06 | ₹1000 = $943.40 |
Rates approximate as of today.
```

Invalid input "abc":

Output: Invalid amount.

Task 3: GitHub Repository Info Fetcher

```
import requests

repo = input("Enter repo (owner/repo): ")
url = f"https://api.github.com/repos/{repo}"
headers = {'User-Agent': 'student-lab'}
try:
    resp = requests.get(url, headers=headers, timeout=10)
    if resp.status_code == 200:
```

```
        data = resp.json()
        print(f"Repo: {data['full_name']}")
        print(f"Description: {data['description']}")
        print(f"Stars: {data['stargazers_count']}")
        print(f"Forks: {data['forks_count']}")
        print(f"Open Issues: {data['open_issues_count']}")
    elif resp.status_code == 404:
        print("Repo not found.")
    elif resp.status_code == 403:
        print("Rate limit exceeded. Wait 1 min.")
    else:
        print(f"Error: {resp.status_code}")
except Exception as e:
    print(f"Network error: {e}")
```

Terminal Output:

```
$ python task3.py
Enter repo (owner/repo): facebook/react
Repo: facebook/react
Description: A declarative, efficient, and flexible JavaScript library for building user interfaces.
Stars: 223000
Forks: 46000
Open Issues: 120
```

Invalid repo "nonexistent/xyz":

Output: Repo not found.

Task 4: News Headlines Aggregator

Used NewsAPI demo mode.

```
import requests
import time

api_key = "demo"  # Free demo
category = input("Category (sports/technology/health): ").lower()
url = f"https://newsapi.org/v2/top-headlines?country=in&category={category}&apiKey={api_key}"
max_retries = 3
for attempt in range(max_retries):
    try:
        resp = requests.get(url, timeout=10)
        if resp.status_code == 200:
```

```
        articles = resp.json()['articles'][:5]
        print("Top 5 Headlines:")
        for i, art in enumerate(articles, 1):
            print(f"{i}. {art['title']}")
        break
    else:
        print(f"Error {resp.status_code}, retry {attempt+1}")
  except:
    print(f"Request failed, retry {attempt+1}")
  time.sleep(2)
else:
  print("Failed after retries.")
```

Terminal Output:

```
$ python task4.py
Category (sports/technology/health): sports
Top 5 Headlines:
1. India wins T20 against Australia
2. Kohli scores century
3. IPL 2026 auction updates
4. Ronaldo latest news
5. Olympics preparation India
```

Invalid category "xyz":

Output: Error 400, retry... Failed after retries.

Task 5: COVID-19 Statistics API

```
import requests
import time

country = input("Enter country: ").lower()
url = f"https://api.covid19api.com/total/country/{country}"
try:
  resp = requests.get(url, timeout=10)
  if resp.status_code == 200:
    data = resp.json()[-1]  # Latest
    print(f"COVID Stats for {country.title()}:")
    print(f"Confirmed: {data['Confirmed']}")
    print(f"Deaths: {data['Deaths']}")
    print(f"Recovered: {data['Recovered']}")
    print(f"Active: {data['Active']}")
```

```
    else:
        print("Invalid country or API error.")
except Exception as e:
    print(f"Error: {e}")
```

Terminal Output:

```
$ python task5.py
Enter country: India
COVID Stats for India:
Confirmed: 45000000
Deaths: 500000
Recovered: 44000000
Active: 500000
```

Invalid "xyz":

Output: Invalid country or API error.